Mitigating Supplier's ESG Greenwashing through Customer AI Applications: Evidence from Chinese Listed Companies

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Mitigating Supplier’s ESG Greenwashing through Buyers’ AI Applications: Evidence from Chinese Listed Companies

Abstract:Mitigating suppliers' ESG greenwashing behaviors is crucial for preventing responsibility spillover and alleviating stakeholder criticism. This study empirically investigates the governance impact of buyers' artificial intelligence (AI) technology applications on suppliers' Environmental, Social, and Governance (ESG) greenwashing practices. Utilizing supply chain data from Chinese A-share listed companies spanning 2010 to 2023, the research reveals that buyers' AI technology deployment significantly curtails suppliers' greenwashing behaviors. This finding remains robust after a series of sensitivity analyses and endogeneity tests. Mechanism analysis elucidates that AI technology attenuates suppliers' greenwashing behaviors by enhancing information transparency and generating technological spillover effects. Further examination indicates that the supply chain power wielded by buyers and the geographical proximity between buyers and suppliers amplify the governance efficacy of AI technology. This research not only expands the application of stakeholder theory within the supply chain context but also deepens the theoretical understanding of AI technology's role in governing supply chain greenwashing.

Key words: ESG Greenwashing, Artificial Intelligence, suppliers and buyers, spillover effect, supply chain power, geographical proximity

1.Introduction

In recent years, the intensification of climate change, environmental pollution, and associated health concerns has propelled the concept of sustainable development, centered on Environmental, Social, and Governance (ESG) principles, to the forefront of societal consciousness (Yang et al., 2024). To address stakeholders' growing concerns regarding sustainable practices, companies have actively integrated ESG principles into their supply chain operations, aiming to achieve competitive advantages through green transformation (Ana Inês et al., 2023). However, in constructing sustainable supply chains, companies face the challenge of effectively enhancing supply chain transparency to counteract suppliers' greenwashing behaviors (Santos et al., 2024). suppliers' greenwashing refers to the phenomenon whereby suppliers conceal unfavorable environmental information or exaggerate their sustainable efforts to appeal to customers and external stakeholders (Pizzetti et al., 2021). This not only undermines the suppliers' credibility but also exacerbates information asymmetry among supply chain members, impeding green supply chain integration and diminishing sustainable supply chain performance (Santos et al., 2024). Moreover, suppliers' greenwashing has a spillover effect on green responsibility. The public, often unaware of the intricacies of supply chain operations, tends to attribute unethical behaviors such as greenwashing to inadequate oversight by core enterprises (Tang et al., 2023; Pizzetti et al., 2021). This transfer of responsibility not only tarnishes the reputation of enterprises but also escalates supply chain risks and opacity. Consequently, effectively curbing supplier’s ESG greenwashing has become a pressing issue.

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Extant research primarily focuses on the impact of internal corporate governance structures, such as board composition and executive team configurations, and external government regulations and environmental legislation on mitigating greenwashing (Deng et al., 2025; Zhao et al., 2024; Jiang et al., 2024). However, the role of buyers—the most critical stakeholders in the supply chain—in inhibiting suppliers' greenwashing has not been thoroughly explored (Ana Inês et al., 2023). Studies have demonstrated that customer leadership styles (e.g., ethical leadership and authoritarian leadership) significantly influence suppliers' green advocacy and greenwashing behaviors (Blome et al., 2017). Additionally, some research indicates that information sharing between buyers and suppliers can effectively curtail suppliers' greenwashing (Chen and Duan, 2023). Nevertheless, due to the diverse interests and information asymmetry among supply chain members, and the insufficient supply chain power of customer enterprises, customer enterprises often struggle to govern suppliers' greenwashing effectively.

In recent times, the significant potential of artificial intelligence technology in promoting sustainable development has garnered widespread academic attention. The deployment of intelligent technology and devices can enhance resource allocation efficiency, improve green total factor productivity, and boost green innovation performance (Chang et al., 2003; Feng et al., 2024). The utilization of industrial robots can ameliorate the working environment, and human-machine collaboration can increase work efficiency and facilitate work-life balance. Leveraging machine learning algorithms can aid in selecting more capable directors and enhance corporate governance capabilities (Zhang and Yang, 2024). Beyond improving corporate ESG performance, AI technology also helps mitigate corporate greenwashing behaviors (Zhang, 2024). By employing technologies such as machine learning for company report analysis, disclosure data examination, and ESG investment data acquisition, companies can reduce information asymmetry between themselves and external stakeholders, thereby curbing greenwashing behaviors (Li et al., 2024). Although existing research has preliminarily explored the role of AI in green transformation, studies on how AI influences supplier green behaviors through supply chain governance effects remain scarce. This paper aims to address this gap and answer the following questions: (1) Can the adoption of AI technology by customer enterprises inhibit suppliers' ESG greenwashing behaviors? (2) How do the supply chain power held by buyers and the geographical distance to suppliers affect the effectiveness of AI governance?

In summary, this study, grounded in stakeholder theory and from the perspective of buyer enterprises, explores the governance role of buyers' AI technology applications on suppliers' ESG greenwashing behaviors. Stakeholder theory posits that influential stakeholders can drive supply chain partners to heed their demands, providing a valuable perspective on how customers can exert influence to reduce dishonest behaviors in ESG disclosures by suppliers. Utilizing actual supply chain samples from A-share listed companies in China from 2010 to 2023, this study empirically examines the inhibitory effect of customer AI applications on suppliers' ESG greenwashing behaviors. After a series of robustness and endogeneity tests, the study's findings

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remain valid. This paper also analyzes the mechanisms through which customer AI transformation influences suppliers' ESG greenwashing behaviors. Customer AI technology applications reduce suppliers' ESG greenwashing through two mechanisms: enhancing supplier information transparency and the technology spillover effect. Additionally, a moderation effect analysis was conducted to delineate the scope of the effect, demonstrating that the governance effect of AI technology is amplified when buyers possess strong supply chain power and when suppliers are geographically proximate.

The contributions of this study are manifold: First, it extends the application of stakeholder theory in supply chains. Although stakeholder theory is frequently employed in research on sustainable supply chains, few studies have incorporated AI technology applications into the research framework. As AI technology rapidly permeates, exploring the spillover effects of customer AI technology applications on supply chain members' greenwashing behaviors based on stakeholder theory holds significant importance. Second, it enriches the research on AI technology and sustainable development. Existing literature explores the value of AI technology applications in enhancing corporate sustainable performance and curbing corporate greenwashing behaviors at the enterprise level. This study further extends the inhibitory effect of AI on greenwashing behaviors from the enterprise level to the supply chain level, enriching the literature on using emerging digital technologies like AI to govern greenwashing behaviors. Third, it elucidates the mechanisms through which customer AI technology applications mitigate suppliers' greenwashing behaviors, encompassing two dimensions: enhancing supplier information transparency to curb greenwashing motives and improving supplier sustainable performance through technology spillover.

2.Literature Review

2.1 ESG greenwashing and the application of AI technology

The concept of greenwashing, initially describing companies' exaggeration of their environmental practices or product eco-friendliness to mask unmet green promises (Yang et al., 2020). As stakeholders increasingly prioritize sustainability, scholars have broadened the concept to encompass ESG (Environmental, Social, and Governance) domains, defining the gap between a company's ESG claims and actual performance as "ESG greenwashing" (Long et al., 2024). Specifically, greenwashing companies disseminate ESG information through deliberate or misleading statements, creating a false green image and reputation. This behavior not only damages the company's overall reputation and diminishes its performance but also fosters stakeholder distrust, reducing consumer purchase intentions and external investor willingness to invest (Zioło et al., 2024).

Artificial Intelligence technology aims to emulate human cognitive processes and intelligent behavior through techniques such as artificial neural networks, natural language processing, computer vision, and robotics (Mariani et al., 2024). With its powerful capabilities in prediction,

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decision-making, and content generation, AI technology is driving significant transformations in various aspects of corporate operations, research and development, and sales (Loureiro et al., 2021). AI can enhance total factor productivity, foster employee innovation, improve the quality of corporate innovation, and ultimately increase corporate performance and market value. Beyond economic value, scholars have also noted AI's potential in promoting corporate sustainability and enhancing ESG performance. Specifically, AI technology can optimize resource allocation, improve a company's green total factor productivity, and drive green technological innovation, thereby enhancing corporate environmental performance (Chang et al., 2003; Feng et al., 2024). Additionally, AI accelerates the automation of business processes, helping employees escape monotonous and repetitive tasks, improving working conditions, and enhancing corporate social performance (Raisch and Krakowski, 2021). AI also strengthens the ability to process large volumes of high-dimensional data, helping to alleviate information asymmetry between shareholders and management, reduce supervision and governance costs, and improve corporate governance performance (Chen et al., 2024).

Recent studies have also found that AI technology helps curb corporate greenwashing behavior. By improving information symmetry between companies and external stakeholders, alleviating corporate financing constraints, and enhancing corporate green innovation capabilities, AI helps mitigate the occurrence of greenwashing behavior (Zhang, 2024; Li et al., 2024).

2.2 Stakeholder theory, customer, and supplier's greenwashing governance

Stakeholder theory is a significant framework in corporate governance and strategy, positing that stakeholders such as buyers, suppliers, employees, public sectors, and local communities can significantly influence organizational operations (Freeman, 1984; Silvestre et al., 2018). Stakeholder theory is widely applied in corporate sustainability to elucidate the external drivers of corporate green practices. Studies reveal the critical role of factors such as stakeholder legitimacy and power status in influencing stakeholder impact (Raha et al., 2021). Among external stakeholders, buyers share risks and benefits with companies, motivating suppliers to improve green behavior and implement sustainable development.

The literature on green supply chain management and supply chain sustainability has long focused on the adverse impact of unethical supplier behavior on the core company's reputation and performance (Blome et al., 2017; Reuter et al., 2010). Suppliers' greenwashing not only hinders supply chain collaboration and integration but also weakens the customer's corporate sustainability performance (Santos et al., 2024). Companies face pressure from external stakeholders such as the media, non-governmental organizations, and governments, demanding accountability for greenwashing behavior within their supply chains (Tang et al., 2023a; Pizzetti et al., 2021), especially when companies fail to take effective measures to monitor suppliers. Therefore, considering the liability spillover effect of suppliers' greenwashing behavior, buyers, as crucial external stakeholders, must pay attention and take relevant measures to govern suppliers'

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greenwashing behavior.

Research on formal contract governance indicates that customer companies can effectively prevent greenwashing issues by collaborating with suppliers and promoting sustainable standards upstream in the supply chain (Blome et al., 2017). From the perspective of relational governance, long-established trust relationships between supply chain members facilitate the dissemination of sustainable development practices and promote sustainability throughout the supply chain (Rajesh, 2023). Trust also fosters information sharing among supply chain members, reducing information asymmetry regarding ESG performance and mitigating the negative effects of suppliers' greenwashing behavior (Chen and Duan, 2023).

Although stakeholders require companies to bear responsibility for suppliers' greenwashing behavior, buyers face numerous challenges in effectively governing such behavior due to information asymmetry caused by suppliers intentionally concealing their true ESG performance, lack of power, and geographic dispersion of suppliers.

As an emerging disruptive digital technology, AI exhibits tremendous potential in enhancing corporate ESG performance and curbing greenwashing behavior. However, there is currently a lack of sufficient research investigating whether AI can play a similar governance role among supply chain members to curb suppliers' ESG greenwashing. Therefore, this paper aims to conduct an in-depth theoretical analysis and empirical examination of this issue.

3.Research Hypotheses

Stakeholder theory posits that pressure from supply chain stakeholders can compel firms to enhance their environmental performance and mitigate unethical behaviors such as greenwashing (Gualandris et al., 2015). In the era of artificial intelligence, the integration of digital technologies, including AI, is becoming increasingly pervasive within supply chains. On one hand, AI enables buyers to comprehensively access information about supply chain operations, carbon emissions, and governance, thereby reducing information asymmetry within the supply chain (Mhlanga, 2021; Modgil et al., 2022). On the other hand, the adoption of AI technology by buyerscan create a technology spillover effect to suppliers, where buyers' sustainable practices, such as ESG initiatives, diffuse upstream in the supply chain, thereby enhancing suppliers' sustainable performance (Wang et al., 2024; Zhang et al., 2024; Li, 2024). Consequently, this study hypothesizes that the adoption of AI technology by buyers significantly influences suppliers' greenwashing behaviors.

Specifically, the governance effect of customer AI technology on suppliers' greenwashing manifests in two key aspects. First, AI technology enhances buyers' capacity to monitor suppliers' greenwashing activities. With AI technology, buyers can collect and analyze various forms of unstructured information to dynamically assess suppliers (Sharma & George, 2021). Leveraging machine learning and natural language processing technologies, firms can scrutinize suppliers' annual reports, news articles, social responsibility reports, and social media data to identify

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potential inconsistencies and anomalous patterns (Khan et al., 2023). The application of AI technology facilitates real-time communication and data sharing between firms and suppliers, aiding in the automated monitoring of suppliers' ESG performance and mitigating greenwashing behaviors stemming from ESG information disclosure (Weisz et al., 2025).

Second, the spillover effect induced by buyers' AI technology applications diminishes suppliers' motivation to engage in greenwashing. According to the diffusion of innovation theory, when buyersadopt AI technology, they provide suppliers with opportunities to access and learn emerging technologies, leading to the dissemination of AI within the supply chain. This fosters improvements in suppliers' ESG performance and reduces their propensity for greenwashing behaviors (Wang et al., 2024). Additionally, buyers' AI adoption enhances their own green innovation and resource utilization efficiency. This green innovation effect also permeates the supply chain, further boosting suppliers' green innovation capabilities and reducing the incidence of greenwashing behaviors (Feng et al., 2022). Based on the aforementioned arguments, the following hypothesis is proposed:

H1: The adoption of artificial intelligence by buyers can significantly reduce the occurrence of ESG greenwashing behaviors among suppliers.

According to signaling theory, information asymmetry between enterprises affects their behaviors (Connelly et al., 2011). In the supply chain, buyers make decisions based on the information conveyed by suppliers. Information asymmetry between firms and their supplier partners can influence the decision-making process (Song et al., 2023). Suppliers might convey misleading environmental information, exaggerating their actual ESG performance to alleviate pressure from stakeholders (Pizzetti et al., 2021). Therefore, suppliers' greenwashing can be conceptualized as misleading information transmission within the supply chain due to information asymmetry. Buyers can effectively mitigate information asymmetry in the supply chain through AI technology, thereby reducing the likelihood of suppliers conveying misleading ESG information.

AI technology enhances information transparency in the supply chain in several ways. First, AI technology promotes the development of the supply chain towards networking and platformization, reducing the time lag in information transmission between firms and suppliers and enhancing real-time information sharing (Sharma et al., 2022). AI-enabled smart connectivity platforms allow buyers to more efficiently supervise suppliers' performance in environmental protection, employee working conditions, and other areas, effectively increasing the transparency of ESG-related information (Birkel and Müller, 2021; Karmaker et al., 2023). The reduction in information transfer steps also helps to decrease distortions caused by intermediary information exchanges.

Second, AI technology enhances buyers' capacity to process information. With technologies such as natural language processing and convolutional neural networks, firms can batch process unstructured information from suppliers, including text, images, and videos, converting them into

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structured and standardized data (Sharma and George, 2021; Rane et al., 2021). Utilizing these technologies, firms can dynamically assess the risk of suppliers' greenwashing and issue warnings about improper supplier behaviors, preventing the escalation of greenwashing and the spillover of responsibility.

Lastly, AI technology allows firms to improve supplier selection models through optimization algorithms, enabling more precise supplier selection and excluding those with poor sustainability performance (Guida et al., 2023). This screening mechanism also incentivizes existing suppliers to avoid potential greenwashing behaviors. In summary, AI technology enhances information transparency in the supply chain, reducing the likelihood of suppliers conveying false ESG signals. Based on the aforementioned arguments, the following hypothesis is proposed:

H2: The application of AI technology by buyers reduces suppliers' ESG greenwashing behaviors by enhancing information transparency in the supply chain.

According to the theory of innovation diffusion, novel technologies developed by innovators are subsequently adopted and disseminated by adopters through learning and imitation processes (Rogers et al., 1962; Stoneman, 1983). This theory posits that technology diffusion is a selective process wherein firms prioritize the adoption of more efficient, cost-effective, and innovative technologies based on their technical knowledge base and strategic needs (Valente & Davis, 1999). In the era of supply chain competition, collaboration between upstream and downstream suppliers has become increasingly intensive. The experience and knowledge accumulated by buyersin the application of artificial intelligence technology will diffuse throughout the supply chain (Wang et al., 2024).

On one hand, the diffusion of AI technology helps inhibit greenwashing behavior among suppliers. As a general-purpose technology, AI brings significant changes to business operations and models. The competitive advantages achieved by buyers through AI applications, such as improved operational efficiency, enhanced decision-making quality, and superior customer experience (Hossain et al., 2022), provide a demonstration effect for suppliers. When suppliers observe the competitive advantages conferred by AI technology, they are likely to adopt this technology proactively (Wang et al., 2024). By implementing AI technology, suppliers can effectively reduce resource consumption and pollutant emissions, thereby improving their actual ESG performance and mitigating greenwashing behavior caused by inaccurate ESG information disclosure (Chang et al., 2003; Feng et al., 2024).

On the other hand, the diffusion of green technology within the supply chain reduces greenwashing behavior among suppliers. The application of AI technology by buyers accelerates the diffusion of green technology within the supply chain. Buyers utilize reinforcement learning models to predict energy consumption ratios of green products, thereby shortening new product development cycles (Yang et al., 2022). They also employ deep learning models to analyze green process technology parameters, iterating and updating industrial processes to enhance green

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innovation capabilities. As buyers' green innovation capabilities improve, they impose stricter environmental standards on suppliers to ensure the overall green performance of products (Kinkel et al., 2022). Under such pressure, suppliers may increase R&D investments to bridge the green technology gap. Additionally, when suppliers engage in greenwashing, the public demands that companies bear joint responsibility. Consequently, buyers are motivated to provide green production techniques to suppliers to enhance their green innovation levels (Li, 2024). To improve supplier sustainability, buyers prefer to share green technology with suppliers through R&D cooperation agreements (Wei et al., 2023). Furthermore, buyers accelerate the diffusion of green technology from buyers to suppliers by adopting green information-sharing systems (Khan et al., 2021; Lerman et al., 2022). The enhancement of suppliers' green innovation capabilities can reduce energy consumption during production processes and decrease reliance on fossil fuels, thereby improving environmental quality (Wang et al., 2023). These efforts help enterprises achieve high-quality ESG performance, ultimately reducing the risk of greenwashing among suppliers.

In summary, the technological diffusion effect generated by buyers' AI technology applications can stimulate the adoption of AI technology and the enhancement of green innovation capabilities among suppliers. Through these improvements, suppliers' performance in environmental, social, and corporate governance aspects is enhanced. Therefore, we propose Hypothesis H3:

H3: Buyers' AI technology applications reduce suppliers' ESG greenwashing behavior through technological spillover effects.

In supply chains, power is often defined as the ability to influence others' behavior to achieve desired outcomes (Fan, 2022). Firms with strong supply chain power can leverage key resources such as brand and core technologies to control market entry opportunities for supply chain partners (Cox, 2001). These partners must rely on such resources to enter the market. Consequently, the party at a dependency disadvantage must actively respond to the concerns of the more powerful firm and strive to establish long-term cooperative relationships (Ahmed & Shafiq, 2022).

Based on this logic, this paper posits that the supply chain power held by buyers can enhance the inhibitory effect of buyers' AI technology applications on suppliers' greenwashing. On one hand, customer power helps suppliers share their true ESG performance information. Firms with strong power have greater influence over suppliers, prompting them to integrate into the customer-led supply chain digital platforms, thereby sharing operational and sustainability-related information (Tian et al., 2025). With access to genuine ESG information from buyers, AI technology can more effectively analyze suppliers' greenwashing behavior and issue timely warnings.

On the other hand, the power held by buyers facilitates the spillover effects of AI and green technology. Firms with power often set benchmarks after adopting AI or green technologies,

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triggering imitation among suppliers (Huo et al., 2022). Research has shown that firms' supply chain power can amplify the diffusion of their sustainability initiatives within the supply chain (Tian et al., 2025). However, when buyers' power is relatively low, it becomes more challenging to obtain genuine ESG performance information from suppliers, and the spillover of relevant technologies is also hindered. In summary, we propose Hypothesis H4:

H4: Buyers' supply chain power enhances the governance effect of buyers' AI technology applications on suppliers' ESG greenwashing.

The dissemination of information and the spillover of technological knowledge are both influenced by geographical distance. Economic geography research shows that geographical proximity facilitates the rapid flow of technological knowledge between firms, as knowledge spillover effects diminish with increasing geographical distance（Thompson and Fox-Kean, 2005）. The geographical proximity between suppliers and buyers facilitates frequent interactions and communications, establishing trust relationships that are crucial for the dissemination of tacit knowledge（Isaksson et al., 2016）. Existing studies have also found that geographical proximity between firms and suppliers facilitates the spillover of digital and green technologies (Li, 2024). Moreover, ESG-related information, being subjective and soft in nature, makes it difficult for firms to directly assess the discrepancies between the information disclosed by suppliers and the actual information（Capelle-Blancard and Petit, 2019; Hasan et al., 2022）. In such cases, face-to-face communication and on-site visits become key to obtaining genuine ESG performance from suppliers. Therefore, geographical proximity of suppliers facilitates frequent face-to-face communication and on-site visits, thereby enhancing firms' ability to grasp suppliers' true ESG performance（Cao et al., 2024）. Conversely, the greater the geographical distance among supplier members, the harder it is for firms to obtain genuine ESG performance information from suppliers. The supervision and regulation of suppliers' greenwashing behavior using AI technology are based on a large amount of information input, and insufficient information reduces the effectiveness of firms' AI technology. In conclusion, we propose the following hypothesis:

H5: Geographical proximity enhances the inhibitory effect of buyers' AI technology applications on suppliers' greenwashing.

4. Methods

4.1 Data source

This study utilizes a panel dataset of Chinese listed companies from 2010 to 2023. Following the methodology of Chen and Xu (2023), we constructed a customer-supplier-year dataset. By leveraging the top five customer lists of listed companies available in the China Securities Market and Accounting Research (CSMAR) database, we matched suppliers with their corporate buyers, resulting in a dyadic customer-supplier-year database. We excluded: (1) samples where the top five buyers do not include listed companies; (2) ST, *ST, and PT companies with abnormal operations; (3) financial and real estate companies; and (4) samples with missing key variables. To

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mitigate the impact of outliers, all continuous variables were winsorized at the 1% and 99% levels. The final sample consists of 997 supplier-buyer-year observations. ESG data were obtained from the WINDS database, AI usage data from annual reports of listed companies, and other financial data from the CSMAR database.

4.2 Model specification

To investigate the mitigating effect of customer AI applications on suppliers' greenwashing, we specify the following regression model:

$$ESGWS_{it} = \beta_0 + \beta_1 CustAI_{it} + \beta_2 Controls_{it} + \beta_3 Industry_i + \beta_4 Year_t + \varepsilon_{it} \quad (1)$$

where i denotes the firm and $t$denotes the year. $ESGWS$ represents the degree of greenwashing by supplier， $CustAI$represents the AI applications by buyers, and $Controls$ represents a vector of control variables influencing suppliers' greenwashing. $Industry$ and $Year$ denote industry and year fixed effects, respectively.

Building on Yang et al. (2023), we construct the following mediation models to explore the mechanisms through which customer AI applications impact suppliers' greenwashing behavior:

$$M_{it} = \alpha_0 + \alpha_1 CustAI_{it} + \alpha_2 Controls_{it} + \alpha_3 Industry_i + \alpha_4 Year_t + \varepsilon_{it} \quad (2)$$

$$ESGWS_{it} = \gamma_0 + \gamma_1 CustAI_{it} + \gamma_2 M_{it} + \gamma_3 Controls_{it} + \gamma_4 Industry_i + \gamma_5 Year_t + \varepsilon_{it} \quad (3)$$

In model (2), M_{it} represents the mediating variables related to information transparency and technology spillover. The remaining variables are consistent with model (1). A mediation effect is indicated when β_1 α_1, and $\gamma_2$are all significant, and $\beta_1 > \alpha_1$.

To examine the moderating effects of supply chain power and geographical distance on suppliers, we specify the following model:

$$ESGWS_{it} = \varphi_0 + \varphi_1 CustAI_{it} + \varphi_2 Mode_{it} * CustAI_{it} + \varphi_3 Controls_{it} + \varphi_4 Industry_i + \varphi_5 Year_t + \varepsilon_{it} \quad (4)$$

In model (4),$Mode_{it}$ represent the moderating variables of supply chain power and geographical distance, respectively. The remaining variables are consistent with model (1).

4.3 Variable measurement

4.3.1 Dependent variable

ESG greenwashing is defined as the discrepancy between a firm's actual ESG performance and its disclosed ESG information. This study employs the methodology proposed by Long et al. (2024) to operationalize ESG greenwashing, utilizing the following measurement procedure:

Step 1: Define the actual ESG performance of enterprises and the status of their ESG information disclosure. The Huazheng (China Securities) ESG rating is used as a proxy for actual ESG performance, while the Bloomberg ESG rating measures ESG information disclosure. The Huazheng ESG rating system employs a dynamic indicator framework tailored to the Chinese market, incorporating both structured and unstructured data to evaluate firms' ESG performance,

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thus providing a more accurate approximation of actual ESG performance. Conversely, the Bloomberg ESG score primarily assesses the transparency of firms' ESG disclosures (Tamimi and Sebastianelli, 2017), rather than directly reflecting actual ESG performance.

Step 2: Analyze the divergence between firms' actual ESG performance and their ESG disclosure scores. To ensure comparability across different evaluation systems, both the actual ESG scores and the ESG disclosure scores are standardized beforehand.

Step 3: Quantify suppliers' ESG greenwashing(Esgws). The extent of ESG greenwashing for a given firm is calculated using the following formula:

$$ESGWS_{it} = \frac{ESG_{ed} - \overline{ESG_{ed}}}{\sigma_d} - \frac{ESG_{ep} - \overline{ESG_{ep}}}{\sigma_p} \quad (5)$$

where ESG_{ed} andESG_{ep}denote the Bloomberg and Huazheng ESG scores for firmi, respectively. $\overline{ESG_{ed}}$ and $\overline{ESG_{ep}}$ represent the mean Bloomberg and Huazheng ESG scores within the industry, respectively. σ_d and σ_p signify the standard deviation of the Bloomberg and Huazheng ESG scores within the industry, respectively.

4.3.2 Independent variable

Building on prior research (Li et al., 2024; Chen et al., 2024), this study constructs a proxy for buyers' artificial intelligence technology adoption(CustAI) by analyzing the frequency of AI-related terms in the annual reports of listed companies. Initially, we gathered the annual reports of these companies. Next, using the approach proposed by Mikolo et al. (2013), we employed the SKIP-gram model to develop a lexicon indicative of AI technology application, resulting in a list of 62 terms. We then utilized Python for word segmentation and frequency analysis of AI-related terms in the annual reports. Finally, the natural logarithm of the AI keyword frequency plus one was used as a proxy for corporate AI usage.

4.3.3 Moderating variables

Supply Chain Power ($Power$):According to resource dependence theory, power arises from asymmetric dependence on supply chain partners. A high concentration of suppliers suggests increased dependence on those suppliers due to fewer alternatives, leading to reduced supply chain power, and vice versa. Consequently, we use the top five supplier concentration of buyersas a proxy for supply chain power. Following Fan et al. (2024), we use 1 minus the top five supplier concentration as a substitute variable for supply chain power.

Supplier Geographical Proximity($Dist$) : Drawing on Kang and Kim (2008), we first calculate the geographical distance between suppliers using the Euclidean distance between the registration points of the firm and its suppliers, as detailed in Equation (6):

$$geo_{ij} = R \times arccos \left[\sin(lat_i) \times \sin(lat_j) + \cos\left(lat_i\right) \times \cos lat_j \times \cos\left(|long_i - long_j|\right)\right] \quad (6)$$

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where lat_i andlat_j represent the latitudes of organizations $i$and j, respectively, and $long_i$ and $long_j$represent their longitudes. The constant 6371 is the Earth's radius in kilometers.

Subsequently, following Cao et al. (2024), we calculate supplier geographical proximity using Equation (7):

$$Dist_{ij} = \frac{maxgeo_{ij} - geo_{ij}}{maxgeo_{ij} - mingeo_{ij}} \quad (7)$$

4.3.4 Mechanism variables

Information Transparency(IC): The adoption of artificial intelligence technology by buyers may enhance supply chain information transparency, potentially mitigating information asymmetry between firms and suppliers, thus inhibiting suppliers' greenwashing behavior. This study primarily focuses on supplier firm information transparency, utilizing the suppliers information disclosure evaluation results from the Shenzhen and Shanghai Stock Exchanges as a proxy, consistent with Li et al. (2022). These exchanges assess listed companies' internal information disclosure, categorizing results into four grades: A, B, C, and D. We assign scores of 4, 3, 2, and 1 respectively to these grades, with higher scores indicating greater supply chain information transparency (IC).

Technology Spillover:Customer AI technology adoption may enhance suppliers' AI application and green innovation capabilities through technology spillover effects, potentially suppressing suppliers' greenwashing behavior. We measure this spillover effect through two dimensions: AI technology spillover and green innovation capability spillover. Supplier AI technology application (SupplAI) is quantified by the word frequency of supplier AI mentions (measured similarly to customer firms). Supplier green innovation capability (SupplGI) is measured by the number of green patent applications filed by suppliers.

4.3.5 Control variables

Drawing on extant literature, we control for supplier firm characteristics that may influence greenwashing (Li, 2024). These include:Firm size (Size): natural logarithm of total assets.Leverage (Lev): ratio of total liabilities to total assets.Firm performance (Roa): return on total assets.Growth(Growt$h$): operating income growth rate.Board size （Board）:the number of board members.Independent director ratio (Director):the ratio of independent directors to total directors in the company.Ownership concentration ($Sh1$): largest shareholder's ownership percentage.CEO duality （Dual）: whether CEO and chairman roles are combined. Cash flow position （Cas$h$）: cash flow to total liabilities ratio.Ownership nature （Soe）: state-owned enterprise indicator.

4.4 Descriptive statistics

Table 1 presents descriptive statistics for the main variables. The dependent variable, greenwashing, has a mean of -0.377 (SD = 1.278), ranging from -3.588 to 3.564. The independent variable, customer firm AI usage, averages 0.213 (SD = 0.537), ranging from 0 to 3.296. Supplier

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firm AI usage averages 0.193 (SD = 0.541), suggesting buyerslead in AI adoption. Supplier green innovation averages 1.741 (SD = 1.733), indicating substantial emphasis on green technology innovation among sample firms. Regarding moderating variables, supply chain power averages 0.335 (SD = 0.201), revealing considerable variation in customer firms' supply chain influence. Geographical distance averages 0.951 (SD = 0.060).

Table 1 Descriptive statistics

Variables	N	Mean	SD	Min	Max
Esgws	979	-0.377	1.278	-3.588	3.564
CustAI	979	0.213	0.537	0	3.296
Power	979	0.335	0.201	0.102	0.990
Dist	979	0.829	0.189	0	1
IC	979	1.931	0.681	1	4
SupplAI	979	0.193	0.541	0	3.998
SupplGI	979	1.741	1.733	0	6.931
Size	979	24.125	1.711	20.184	30.995
Lev	979	0.514	0.185	0.042	1
Roa	979	0.041	0.061	-0.291	0.439
Growth	979	0.122	0.424	-0.862	4.775
Board	979	2.262	0.233	1	2.890
Director	979	0.369	0.073	0.03	2
Sh	979	0.441	0.178	0.02	0.899
Dual	979	0.172	0.378	0	1
Cash	979	0.063	0.067	-0.221	0.371
Soe	979	0.701	0.458	0	1

5. Analysis of Regression Results

5.1 Baseline regression results

Table 2 presents the baseline regression results of this study. Column (1) includes only the independent variables, while Columns (2) to (4) progressively incorporate control variables, as well as time and industry fixed effects. The results indicate that the regression coefficient for customer AI application is negative and statistically significant at the 1% level across all specifications. This finding suggests that the adoption of AI technology by buyers is associated with a reduction in suppliers' ESG greenwashing behavior, thus supporting Hypothesis H1.

Table 2 Baseline regression results

	（1）	（2）	（3）	（4）

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Variables	*Esgws*	*Esgws*	*Esgws*	*Esgws*
CustAI	-0.389***	-0.324***	-0.380***	-0.367***
	（0.075）	(0.073)	（0.075）	(0.076)
Size		0.191***	0.146***	0.147***
		（0.029）	（0.031）	（0.031）
Lev		-0.585**	-0.460*	-0.380
		（0.263）	（0.263）	（0.270）
Roa		-1.267	-1.097	-0.803
		（0.890）	（0.903）	（0.905）
Growth		1.033	0.047	0.004
		（0.097）	(0.099)	（0.103）
Board		0.291	0.521***	0.601***
		（0.193）	(0.200)	（0.213）
Director		0.290	0.571	0.189
		（0.556）	(0.555)	（0.555）
Sh		-0.001	0.001	-0.002
		（0.002）	(0.002)	（0.002）
Dual		0.136	0.153	0.049
		（0.111）	(0.111)	（0.117）
Cash		0.103	-0.394	-0.258
		（0.692）	(0.701)	（0.707）
Soe		0.196**	0.185*	0.113
		（0.099）	(0.099)	（0.104）
Constant	-0.294***	-5.427***	-0.530***	-6.875***
	（0.043）	（0.675）	(0.693)	（1.400）
Year Fixed	No	No	Yes	Yes
Industry Fixed	No	No	No	Yes
N	979	979	979	979
Adjusted R^2	0.025	0.093	0.109	0.182

Note: Cluster-robust standard errors are presented in parentheses. The symbols ***, **, and * denote statistical significance of the estimated coefficients at the 1%, 5%, and 10% levels, respectively.

5.2 Robustness checks

5.2.1 Alternative measurement of independent variable

While numerous studies commonly employ the frequency of AI-related terms in the management discussion section of annual reports to measure firms' use of AI technology, this metric primarily reflects the extent of public disclosure rather than actual application. In contrast, data on AI patent applications more accurately capture a firm's tangible investment and deployment in AI-related technologies. Therefore, we utilize the number of customer AI patent applications as an alternative proxy for customer AI technology application (`CustAIpatient`). The results of this analysis, presented in Column (1) of Table 3, reveal a statistically significant

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negative regression coefficient for the number of customer AI patent applications, providing further support for Hypothesis H1.

Table 3 Robustness test

	(1)	(2)	(3)	(4)
	Alternative measurement of AI	Buyers' controls	Regional FE	Lagged variables
Variables	Esgws	Esgws	Esgws	Esgws
CustAI		-0.390***	-0.378***	
		(0.078)	(0.079)	
CustAIpatient	-1.271***			
	(0.039)			
$CustAI_{t-1}$				-0.208**
				(0.084)
Size	0.155***	0.148***	0.129***	0.154***
	(0.039)	(0.032)	(0.034)	(0.031)
Lev	-0.374	-0.402	-0.377	-0.381
	(0.272)	(0.273)	(0.292)	(0.272)
Roa	-0.978	-0.102	-0.782	-0.883
	(0.912)	(0.913)	(0.926)	(0.914)
Growth	0.002	-0.016	-0.002	-0.017
	(0.103	(0.103)	(0.104)	(0.103)
Board	0.651***	0.588***	0.452***	0.599***
	(0.215)	(0.217)	(0.223)	(0.216)
Director	0.241	0.235	0.088	0.181
	(0.559)	(0.564)	(0.569)	(0.560)
Sh	-0.002	-0.001	-0.002	-0.002
	(0.002)	(0.002)	(0.003)	(0.002)
Dual	0.015	0.047	-0.019	0.025
	(0.118)	(0.119)	(0.123)	(0.118)
Cash	-0.094	-0.133	0.027	-0.157
	(0.712)	(0.714)	(0.724)	(0.713)
Soe	0.136	0.146	0.142	0.131
	(0.105)	(0.105)	(0.113)	(0.105)
Constant	-7.890***	-0.709***	-7.495***	-7.558***
	(1.401)	(1.570)	(1.884)	(1.403)
Year Fixed	Yes	Yes	Yes	Yes
Industry Fixed	Yes	Yes	Yes	Yes
Customer Characteristics	No	No	Yes	No
Regional Fixed	No	Yes	No	No
N	979	979	979	979
Adjusted R^2	0.171	0.182	0.187	0.167

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Note: Cluster-robust standard errors are presented in parentheses. The symbols ***, **, and * denote statistical significance of the estimated coefficients at the 1%, 5%, and 10% levels, respectively.

5.2.2 Inclusion of omitted variables

To mitigate potential endogeneity concerns arising from omitted variables, we extend our baseline model by incorporating customer firm characteristics as additional control variables, calculated using the same methodology as supplier firm characteristics. The results, displayed in Column (2) of Table 3, demonstrate that the regression coefficient for customer AI application remains negative and statistically significant at the 1% level after controlling for customer firm characteristics, consistent with our baseline findings. This robustness check enhances confidence in our study's conclusions.

5.2.3 Controlling for provincial-level fixed effects

To account for the potential impact of omitted macro-regional variables, such as disparities in regional economic development and marketization levels, we introduce provincial-level fixed effects into our model. The results, presented in Column (3) of Table 3, show that the regression coefficient for customer AI application remains negative and statistically significant at the 1% level after controlling for provincial-level fixed effects, further corroborating Hypothesis H1.

5.2.4 Consideration of Lagged Effects

Recognizing that the spillover effects of customer AI technology may exhibit a temporal lag, as suppliers require time to absorb customer knowledge and implement technological upgrades, we re-estimate our model using a one-year lagged measure of customer AI technology application ($CustAI_{t-1}$) as the independent variable. The results, reported in Column (4) of Table 3, reveal that the regression coefficient for the one-year lagged customer AI technology application is negative and statistically significant at the 5% level, providing additional support for Hypothesis H1.

5.3 Endogeneity test

5.3.1 Instrumental variable approach

To address potential endogeneity concerns arising from reverse causality, we employ an instrumental variable (IV) approach. Following Tang et al. (2023), we utilize the mean industry-level artificial intelligence development, excluding the focal firm, as our first instrument ($IV1$). This industry-level AI metric correlates with firm-level AI adoption but is not directly influenced by individual firm behavior, thus satisfying the exogeneity condition.

Additionally, we construct a second instrument ($IV2$) based on the heteroskedasticity-based approach proposed by Lewbel (1997). Specifically, we use the cubic difference between a firm's AI development level and the industry-province average. This instrument is designed to be uncorrelated with suppliers' greenwashing residuals after controlling for fixed effects, while maintaining a strong correlation with firm-level AI adoption.

Table 5 presents the two-stage least squares (2SLS) regression results using these

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instruments. The first-stage F-statistics consistently exceed 10, indicating the relevance of our constructed instruments. The second-stage results reveal significantly positive coefficients for the instrumented variables, indicating that the baseline regression results of this paper remain robust after considering endogeneity problems. The Kleibergen-Paap rk LM statistic rejects the null hypothesis at the 1% level, confirming instrument validity.

Table 4 Endogeneity test

	2SLS estimation with instrumental variables.				PSM
	（1）	（2）	（3）	（4）	(5)
	First Stage	Second Stage	First Stage	Second Stage	n=1
Variables	*CustAI*	*Esgws*	*CustAI*	*Esgws*	*Esgws*
*IV*1	0.134***				
	(0.019)				
*IV*2			0.002**		
			(0.001)		
Cust*AI*		-0.609***		-1.987*	-0.480***
		（0.229）		(0.965)	（0.105）
Size	-0.019	0.140***	-0.027**	0.102**	0.154*
	（0.135）	（0.031）	（0.013）	(0.041)	（0.086）
Lev	-0.001	-0.383	-0.026	-0.320	-0.719
	（0.112）	（0.254）	(0.114)	(1.037)	（0.656）
Roa	0.262	-0.731	0.281	-0.320	-0.483
	（0.299）	（0.902）	(0.326)	(1.037)	（2.304）
Growth	0.037	0.002	0.021	0.040	-0.022
	（0.039）	（0.094）	(0.040)	(1.211)	（0.207）
Board	-0.039	0.584***	-0.063	0.490*	1.034**
	（0.098）	（0.220）	(0.103)	(0.259)	（0.491）
Director	-0.031	0.186	0.015	0.160	2.142
	（0.149）	（0.566）	(0.177)	(0.630)	（2.036）
Sh	0.001	-0.002	-0.001	-0.002	-0.005
	(0.001)	（0.002）	(0.001)	（0.003）	（0.006）
Dual	0.093*	0.075	0.101*	0.224	-0.092
	(0.056)	（0.118）	(0.060)	（0.176）	（0.216）
Cash	-0.254	-0.333	-0.338	-0.764	-0.019
	(0.273)	（0.784）	(0.317)	（0.957）	（1.771）
Soe	-0.032	0.105	-0.030	0.063	-0.121
	(0.049)	（0.104）	(0.052)	（0.136）	（0.204）
Constant	-1.930***	-6.356***	2.115***	-3.401*	-8.510***
	(0.309)	(0.871)	（0.313）	(2.129)	（2.513）
Year Fixed	Yes	Yes	Yes	Yes	Yes
Industry Fixed	Yes	Yes	Yes	Yes	Yes

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N	979	979	979	979	237
Adjusted R^2	0. 171	0. 242	0. 141	0. 177	0. 266
Cragg-Donald Wald F	46. 161		47. 773		
Kleibergen-Paap rk LM		82. 982		283. 871	

Note: Cluster-robust standard errors are presented in parentheses. The symbols ***, **, and * denote statistical significance of the estimated coefficients at the 1%, 5%, and 10% levels, respectively.

5.3.2 Propensity score matching (PSM)

To mitigate potential sample selection bias, such as differences in supplier selection criteria between high and low AI-adopting firms, we employ propensity score matching (PSM). We first dichotomize our sample into AI-adopting (treatment) and non-adopting (control) groups based on firms' AI technology adoption. Using our control variables, we conduct a logit regression to calculate propensity scores. We then perform 1:1 nearest neighbor matching with replacement. Post-matching t-tests reveal no statistically significant differences between matched groups across variables, indicating successful balancing.

We re-estimate Model 1 using the matched sample, with results presented in Column 5 of Table 4. The regression coefficient for buyers' AI adoption remains negative and significant at the 1% level, corroborating the robustness of our findings.

5.4 Mechanism testing and moderation effect analysis

5.4.1 Mechanism testing

Our theoretical framework posits that buyers' AI adoption inhibits suppliers' greenwashing by enhancing supply chain information transparency and through technology spillover effects. We empirically test these mechanisms.

1.Information Effect

To examine the mediating role of supply chain information transparency, we utilize the information disclosure evaluation scores of listed companies published by the Shenzhen and Shanghai Stock Exchanges as our mediating variable（IC）. Table 5 presents the mediation analysis results. Column (1) shows the main effect. Column (2) reveals a significantly positive coefficient for buyers' AI adoption, indicating that buyers' AI adoption enhances supplier information transparency. Column (3) shows significantly negative coefficients for both supplier information transparency and buyers' AI adoption, with the latter coefficient smaller in magnitude than the main effect. This pattern suggests that buyers' AI adoption inhibits suppliers' greenwashing partially through improved supply chain information transparency.

2 Technology Spillover Effect

We posit that buyers' AI adoption enhances suppliers' AI capabilities（$SupplAI$） and green innovation（$SupplGI$）, thereby inhibiting greenwashing. Columns (4) and (6) of Table 5 show significantly positive coefficients for buyers' AI adoption on supplier AI adoption and green

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innovation, respectively. Columns (5) and (7) reveal significantly negative coefficients for supplier AI adoption, green innovation, and buyers' AI adoption, with the latter coefficient smaller than the main effect. These results support the technology spillover mechanism, confirming Hypothesis H3.

Table 5 Mechanism Testing

	(1)	(3)	(4)	(5)	(6)	(7)
Variables	Esgws	Esgws	*SupplAI*	Esgws	*SupplGI*	Esgws
Cust*AI*	-0.367***	-0.354***	0.257***	-0.328***	0.312**	-0.341***
	(0.076)	(0.077)	(0.031)	(0.079)	(0.113)	(0.077)
IC		-0.458*				
		(0.272)				
SupplAI				-0.152**		
				(0.080)		
SupplGI						-0.050**
						(0.022)
Size	0.147***	0.148***	0.041***	0.153***	-0.008	0.146***
	(0.031)	(0.031)	(0.013)	(0.0317)	(0.040)	(0.031)
Lev	-0.380	-0.393	-0.100	-0.395*	0.538*	-0.352
	(0.270)	(0.270)	(0.111)	(0.210)	(0.299)	(0.270)
Roa	-0.803	-0.844	-0.170	-0.829	0.782	-0.763
	(0.905)	(0.905)	(0.374)	(0.904)	(1.340)	(0.904)
Growth	0.004	0.001	-0.032	-0.009	-0.007	-0.005
	(0.103)	(0.102)	(0.042)	(0.102)	(0.152)	(0.102)
Board	0.601***	0.584***	-0.112	0.583***	-0.367*	0.582***
	(0.213)	(0.213)	(0.088)	(0.213)	(0.216)	(0.213)
Director	0.189	0.160	-0.110	0.172	-0.424	0.168
	(0.555)	(0.555)	(0.229)	(0.554)	(0.821)	(0.554)
Sh	-0.002	-0.001	-0.004***	-0.002	0.002	-0.002
	(0.002)	(0.002)	(0.001)	(0.002)	(0.004)	(0.002)
Dual	0.049	0.050	0.012	0.051	0.119	0.055
	(0.117)	(0.117)	(0.048)	(0.117)	(0.173)	(0.117)
Cash	-0.258	-0.218	-0.431	-0.324	0.201	-0.248
	(0.707)	(0.707)	(0.292)	(0.707)	(1.046)	(0.705)
Soe	0.113	0.121	-0.074*	0.101	0.013	0.113
	(0.104)	(0.104)	(0.043)	(0.104)	(0.154)	(0.104)
Constant	-6.875***	-3.822*	-0.986*	-7.026***	2.903	-6.727***
	(1.400)	(2.346)	(0.578)	(1.400)	(2.071)	(1.393)
Year Fixed	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed	Yes	Yes	Yes	Yes	Yes	Yes
N	979	979	979	979	979	979

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Adjusted R^2	0.182	0. 184	0. 222	0. 184	0. 107	0. 186

Note: Cluster-robust standard errors are presented in parentheses. The symbols ***, **, and * denote statistical significance of the estimated coefficients at the 1%, 5%, and 10% levels, respectively.

5.4.2 Moderation effect analysis

We examine the moderating effects of supply chain power ($Power$) and supplier geographic proximity ($Dist$). Table 6, Column (1) shows a significantly negative coefficient for buyers' AI adoption and its interaction with supply chain power, supporting Hypothesis H4 that buyers' supply chain power enhances the inhibitory effect of AI adoption on suppliers' greenwashing. Column (2) reveals significantly negative coefficients for buyers' AI adoption and its interaction with geographic proximity, supporting Hypothesis H5 that buyer-supplier proximity facilitates the inhibitory effect of buyers' AI adoption on suppliers' greenwashing.

Table 6 Moderation Effects

	(1)	(2)	(3)
Variables	Esgws	Esgws	Esgws
CustAI	-0. 175***	-0. 193***	-0. 175***
	(0. 042)	(0. 041)	(0. 042)
$Power$	0. 147***		0. 150***
	(0. 043)		(0. 043)
$Dist$		0. 074*	0. 074*
		(0. 046)	(0. 045)
Cust$AI*Power$	-0. 118***		-0. 104***
	(0. 041)		(0. 043)
Cust$AI*Dist$		-0. 129***	-0. 100*
		(0. 059)	(0. 061)
Size	0. 153***	0. 143***	0. 149***
	(0. 031)	(0. 031)	(0. 031)
Lev	-0. 291	-0. 376	-0. 279
	(0. 268)	(0. 270)	(0. 269)
Roa	-0. 748	-0. 722	-0. 672
	(0. 897)	(0. 905)	(0. 897)
Growth	-0. 026	-0. 028	-0. 044
	(0. 102)	(0. 103)	(0. 102)
Board	0. 633***	0. 563***	0. 587***
	(0. 211)	(0. 217)	(0. 215)
Director	0. 069	0. 126	0. 024
	(0. 550)	(0. 555)	(0. 550)
Sh	-0. 001	-0. 002	-0. 001
	(0. 002)	(0. 002)	(0. 002)
Dual	0. 011	0. 033	0. 002
	(0. 116)	(0. 117)	(0. 116)

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Cash	−0. 233	−0. 234	−0. 210
	(0. 700)	(0. 706)	(0. 698)
Soe	0. 090	0. 104	0. 077
	(0. 103)	(0. 104)	(0. 104)
Constant	−7. 158***	−6. 822***	−7. 032***
	(1. 397)	(1. 407)	（1. 411）
Year Fixed	Yes	Yes	Yes
Industry Fixed	Yes	Yes	Yes
N	979	979	979
Adjusted R^2	0. 199	0. 255	0. 270

Note: Cluster-robust standard errors are presented in parentheses. The symbols ***, **, and * denote statistical significance of the estimated coefficients at the 1%, 5%, and 10% levels, respectively.

6. Discussion

This study investigates the impact of customer artificial intelligence technology applications on suppliers' greenwashing behavior within the context of supply chain sustainability. Utilizing supply chain data from china A-share listed companies between 2010 and 2023, our empirical analysis yields several pivotal findings. Firstly, we demonstrate that customer AI technology applications significantly mitigate suppliers' greenwashing behavior, a result that remains robust across various sensitivity tests. Secondly, our mechanism analysis reveals that this inhibitory effect is primarily achieved through enhanced information transparency and technological spillover effects. Lastly, we find that the effectiveness of customer AI technology in curbing suppliers' greenwashing is amplified when buyers possess strong supply chain power and when there is geographical proximity between buyers and suppliers.

6.1Theoretical implications

Our study makes several noteworthy contributions to the existing literature. First, we extend the application of stakeholder theory in supply chain management. While prior research has established the role of buyers in promoting supplier sustainability, our findings highlight how AI technology can serve as a powerful tool for buyers to influence supplier unethical behavior. Moreover, the power of customer firms in the supply chain and their geographical proximity to suppliers can significantly enhance the effectiveness of customer artificial intelligence technology in governing suppliers’ greenwashing. This study provides new insights into the influence of customers and other stakeholders in the supply chain.

Second, we provide empirical evidence for the spillover effects of customer AI technology in fostering supply chain sustainability. While previous studies have explored the spillover of customer sustainable practices and the impact of AI on a firm's own ESG performance, our research bridges these streams by demonstrating how customer AI applications can influence the greenwashing behavior of other supply chain members. This contribution advances the literature on supply chain spillover effects and technology diffusion.

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Lastly, we propose and validate a dual mechanism through which customer AI technology inhibits suppliers' greenwashing. Our findings suggest that AI applications not only enhance information transparency, thereby deterring greenwashing, but also facilitate technological spillovers that improve suppliers' sustainable development capabilities. This dual-mechanism framework provides a more nuanced understanding of how digital technologies can shape supply chain sustainability.

6.2 Managerial implications

Our research offers valuable insights for practitioners aiming to build sustainable supply chains. Firstly, we emphasize the potential of AI as an effective tool for downstream firms to control unethical behavior among supply chain partners. By investing in AI technology, focal companies can not only enhance their own ESG reporting transparency but also improve their ability to monitor and verify the ESG performance of their suppliers.

Secondly, our findings underscore the importance of technology diffusion in supply chains. We recommend that buyersactively promote the spillover of AI and green technologies through collaborative R&D initiatives, resource sharing, and technical support programs. Such efforts can enhance suppliers' sustainable development capabilities, reducing their propensity for greenwashing and mitigating potential liability risks for focal firms.

Lastly, our results suggest that firms can amplify the benefits of AI technology by strategically managing their supply chain relationships. Specifically, companies should focus on building strong supply chain power and prioritize geographically proximate suppliers to maximize the effectiveness of their AI-driven sustainability initiatives.

6.3 Limitations and future research

While our study provides valuable insights, several limitations present opportunities for future research. Firstly, our measurement of AI technology adoption relies primarily on the frequency of AI-related terms in annual reports. Future studies could employ more comprehensive measures, such as survey data or direct indicators of AI investments and human capital, to capture the nuances of corporate AI adoption.Secondly, our focus on the customer perspective leaves room for exploring the reciprocal effects of supplier AI adoption on customer behavior. Future research could investigate whether and how suppliers' AI capabilities influence the sustainability practices of their buyers, providing a more holistic view of AI's role in supply chain sustainability. Lastly, our sample is limited to listed companies and their core listed suppliers. Further research is needed to validate our findings in the context of small and medium-sized enterprises and non-listed suppliers, which may face different constraints and incentives in their sustainability efforts.

In conclusion, our study provides compelling evidence for the role of AI technology in promoting supply chain sustainability and offers a foundation for future research on the

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intersection of digital technologies, corporate governance, and sustainable development in global supply chains.

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